ATDConnect API

Technical Guide

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Publication Information

Publication Date

This document was last published on 03/11/16. The document control number for this document is ATDConnect_3_3_0. The latest ATDConnect version is 3.2.

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1. ATDConnect API Introduction & Overview

1.1 Introduction

ATDConnect is ATD's application programming interface (API). The API provides customer access to the ATD platform through a collection of secure and configurable web services.
Customers can use the API to retrieve real-time, product-inventory information; access product pricing, product availability; and to order products.
ATDConnect 3 is built on a Stateless architecture that provides flexibility, efficient product searches and efficient ordering processes.

1.2 This Document

Audience	This document is written for developers or other technical implementers of the ATDConnect API.
Technical Requirements	The syntax and language used in the document assumes the reader will have working knowledge of the following technologies: • SOAP web services • HTTP • XML/XSD • WSDL • JSON
Document Format	Service methods of this API are detailed in the following format: • Function name • Description • Functional Considerations • WSDL location • Service request parameters • Request Key-Value pairs • Service response parameters • XML payload example(s)

1.3 Glossary of Terms Used in this Document

Term / Acronym	Definition
AMI Number	Automated Media Incorporated (AMI) identification number. This is a unique dealer/location number assigned by AMI for use by new car dealers.
Client	ATDConnect implementer. Typically the person or group that writes the POS software to interface with ATDConnect.
Customer	ATD Tire dealer / vendor
DC	Distribution Center
SOAP	Simple Object Access Protocol – protocol specification for exchanging structured information.
WSDL	Web Services Description Language – xml based language for describing the functionality of web services.
XSD	XML Schema Definition. XML document for specifying the constraints and structure of other documents of that type.

1.4 ATDConnect Web Services Overview

ATDConnect services can be accessed by SOAP requests. The API suite consists of 4 services, and 8 service methods. They are summarized in Table 1.

Table 1: Overview of ATDConnect services

Service	Service Name	Service Method	Description
Dealer Location Service	locations	getLocationByCriteria	Retrieves customer location details. These are unique ship-to locations ('Customer' refers to the dealer)
		getDistributionCenter	Retrieves customer Distribution Center details. Each customer location has 1 assigned Distribution Center
Product Search Service	products	getProductByCriteria	Searches product catalog and retrieves detailed product information using specified search criteria

		getBrand	Retrieves product brands available to the location
Order Service	orders	previewOrder	Submit an order for preview
		placeOrder	Submit final order for processing
Order Status Service	orderStatus	getOrderStatusByCriteria	Searches orders and retrieves basic order header information
		getOrderDetail	Searches orders and retrieves detailed order information

1.4.1.1 SOAP Requests

All SOAP requests are made over HTTP using SSL with SOAP **version 1.1**.

SOAP WSDLs are located at:

Production:	https://ws.atdconnect.com/ws/<version>/<service name>.wsdl
Sandbox/Test:	https://testws.atdconnect.com/ws/<version>/<service name>.wsdl

1.4.1.1.1 SOAP Authentication

ATDConnect uses WS-Security to pass authentication in plain text in the SOAP Header. The services support the ws-security UsernameToken.

All SOAP requests require the following information in the SOAP header:

Table 2: SOAP Required Parameters

Parameter Name	Type	Format	Description
Username	xs:string	Plain text	Username. This is the username supplied by ATD.
Password	xs:string	Plain text	Password. This is the password supplied by ATD.
clientId	xs:string	Plain Text	Client id assigned to ATDConnect API user by ATD.

A typical ATDConnect SOAP envelope with header should resemble the following:

```
<?xml version="1.0"?>
<soapenv:Envelope xmlns:soapenv=http://schemas.xmlsoap.org/soap/envelope/>

  <soapenv:Header>
    <wsse:Security soapenv:mustUnderstand="1" xmlns:wsse="http://docs.oasis-open.org/wss/2004/01/oasis-
200401-wss-wssecurity-secext-1.0.xsd">
      <wsse:UsernameToken atd:clientId="clientId" xmlns:atd="http://api.atdconnect.com/atd">
      <wsse:Username>Username</wsse:Username>
      <wsse:Password Type="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-username-token-
profile-1.0#PasswordText">Password</wsse:Password>
    </wsse:UsernameToken>
    </wsse:Security>
  </soapenv:Header>

   <soap:Body>
     <!--Request payload-->
   </soap:Body>
</soap:Envelope>
```

1.4.1.2 Named Parameters and Key-Value pairs

The API uses two attribute formats to send information to the ATD platform:

1. Named parameters
2. Key-value pairs

Named Parameters

- This format is mainly used for parameters that are common across different services
- These parameters are specified as XML elements in the request. For example, <locationNumber/>

Key-Value pairs

- These are key-attributes and values used when a service transaction has many properties. It would be unsuitable to specify this many keys as named parameters.
- They include keys used in product search request/responses. For example, tire products may have the following attribute keys:
    - MaxRimWidth
    - MaxTirePressure
    - SpeedRating
    - TemperatureRating
- Keys are case sensitive and must be specified exactly as they appear in this document.
- Key-value pairs are specified using the following syntax:

Key-value XML

```
<getServiceMethodRequest>
…
   <someParam>
      <entry>
         <key>ATD-specified key</key>
         <value>some value</key>
      </entry>
      …
   </someParam>
…
</getServiceMethodRequest>
```

1.4.2 Error Handling

ATDConnect will return error messages to the user if it encounters problems processing the service request. Errors will throw **SOAP Faults.**

There are common authentication error conditions that are thrown by all service-method calls.

Table 3: Error messages

Error Condition	Error Message
Invalid Username /password	Invalid Username/Password. Please verify your request
Invalid client ID	Unable to authenticate User's Credentials. Please contact Online Customer Care if you feel that this User Account should have access to ATDConnect
Invalid user credentials for location	The user making this call does not have access to this location.
Invalid payload request	An error occurred processing your request

The error message inside a SOAP fault will occur in the body>Detail element, and will resemble the following:

```
<soap:Envelope    xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
```

```
   <soap:Header/>
   <soap:Body>
     <faultcode>soap:Server</faultcode>
     <faultstring>ATDConnectException</faultstring>
     <detail>
       <message>The user making this call does not have access to this location.
</message>
     </detail>
   </soap:Body>
</soap:Envelope>
```

2. DEALER LOCATION SERVICE

2.1 Description:

The ATDConnect Dealer location service retrieves dealer-specific data that is required to make further API calls. This data includes:

- Customer Location information – numbers, addresses, etc.
- Distribution Center information -- numbers, addresses, etc.

Available response attributes are listed in the web service function calls for this service below.

The **locationNumber** (obtained from this service) is a required value for all subsequent calls to ***Product*** and ***Order*** services.

It is a unique identifier used by ATD to define a single, specific customer shipping location.

2.2 WSDL Location

Production:	https://ws.atdconnect.com/ws/3_2/locations.wsdl
Sandbox/Test:	https://testws.atdconnect.com/ws/3_2/locations.wsdl

2.3 Service Method Calls

2.3.1 Get Dealer Location Information (getLocationByCriteria)

2.3.2 Function Name

getLocationByCriteria

2.3.3 Description:

Retrieves user Location information using the Client Location Number, AMI Number or Store Number.

A valid account and client identification number is required to access this information.

Method will return an error if more than one location matches the search criteria.

For certain clients, it is possible to return all available locations by not specifying any request criteria.

2.3.4 Request Parameters

Request attributes for this method are specified as name-value pairs. See the getLocationByCriteria request keys table below for valid key names.

See the Sample XML Payload for creating this request

Table 4: getLocationByCriteria request parameters

Parameter Name	Type	Required	Example	Description
Criteria	Criteria Type	No		Key-value pairs
XML Criteria Type				
Key	xs:string	No	locationNumber	See Request key table below
Value	xs:string	No	434471	

2.3.5 Request Keys

Table 5: getLocationByCriteria request keys

Key	Value Type	Required	Example	Description
locationNumber	String	No	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location.
storeNumber	String	No	275	Use StoreNumber if LocationNumber and AMINumber are not known. StoreNumber is a customer-specified store number that is used internally by the store's business. This number must be communicated to ATD so it can be loaded into the ATD customer location data

amiNumber	String	No	18575	Use AMINumber if LocationNumber is not known. AMI Number is also known as 'dealer code'. It is a unique number assigned to car dealers by Automated Media Inc. (AMI)

2.3.6 Response Parameters

See the Sample XML Payload Response for xml formatting:

Table 6: getLocationByCriteria response parameters

Parameter Name	Type	Sample / Format	Description
XML LocationType			
locationName	xs:string	Acme Dealer	Dealer location name. This value is set by ATD
locationNumber	xs:string	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location
phoneNumber	xs:string	xxx-xxx-xxxx	
customerNumber	xs:string	910283	The customer number as determined by ATD. One customer may have many locations.
Address	Address Type		Default location address
servicingDC	xs:string	070	3-character unique ID used to identify the ATD Distribution Center that services this location.
XML Address Type			
address1	xs:string	123 Main Street	Address line 1
address2	xs:string	Suite 1	Address line 2
address3	xs:string		Address line 3
City	xs:string	Huntersville	

State	xs:string	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	12345	5-digit U.S. zip code
plus4	xs:string	1234	U.S. zip code extension. Ex, 20001-**0001**

2.3.7 Error Messages

Table 7: getLocationCriteria error messages

Error Condition	Error Message
Empty search values	Please ensure that valid criteria has been specified

2.3.8 Sample XML Payload

- **Request**

```
<getLocationByCriteriaRequest>
  <criteria>
    <entry>
      <key>locationNumber</key> <= valid options are locationNumber,
                                   amiNumber or storeNumber
      <value>434471</value>
    </entry>
  </criteria>
</getLocationByCriteriaRequest>
```

- **Response**

```
<getLocationByCriteriaResponse>
     <locations>
       <location>
         <locationNumber>434471</locationNumber>
         <locationName>CUSTOMER DEMO #275</locationName>
         <phoneNumber>9999999999</phoneNumber>
         <customerNumber>176238</customerNumber>
         <address>
           <address1>55 COMMERCE AVE</address1>
           <city>ALBANY</city>
           <state>NY</state>
           <zipCode>12206</zipCode>
```

```
            </address>
            <servicingDC>275</servicingDC>
          </location>
        </locations>
</getLocationByCriteriaResponse>
```


2.3.9 Get Distribution Center Information (getDistributionCenter)

2.3.10 Function Name

getDistributionCenter

2.3.11 Description:

Function for retrieving detailed Distribution Center information.

Each dealer has at least one Distribution Center that it is associated with.

2.3.12 Request Parameters

Table 8: getDistributionCenter request parameters

Parameter Name	Type	Required	Sample / Format	Description
servicingDC	xs:string	Yes	070	3-character unique ID used to identify the ATD Distribution Center.

See the Sample XML Payload for creating this request

2.3.13 Request Keys

No name-value pairs for this function exist

2.3.14 Response Parameters

See the Sample XML Payload Response on page 16 for xml formatting:

Table 9: getDistributionCenter response parameters

Parameter Name	Type	Sample / Format	Description
XML DistributionCenter type			
servicingDC	xs:string	070	3-character unique ID used to identify the ATD Distribution Center.
description	xs:string	Charlotte NC DC	
phoneNumber	xs:string	704-348-6921	
Address	Address Type		
XML Address Type			
address1	xs:string	123 Main Street	
address2	xs:string	Suite 1	
address3	xs:string		
City	xs:string	Huntersville	
State	xs:string	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	12345	5-digit U.S. zip code
plus4	xs:string	1234	U.S. zip code extension. Ex, 20001-**0001**

2.3.15 Error Messages

Table 10: getDistributionCenter error messages

Error Condition	Error Message
Empty search values	Please ensure that valid criteria has been specified

2.3.16 Sample XML Payload

- **Request**


```xml
<getDistributionCenterRequest>

   <servicingDC>070</servicingDC>

</getDistributionCenterRequest>
```


- **Response**


```xml
<getDistributionCenterResponse>

   <distributionCenter>
      <servicingDC>070</servicingDC>
      <phoneNumber>704-348-6921</phoneNumber>

      <address>
         <address1>123 Distribution Rd</address1>
         <city>Huntersville</city>
         <state>NC</state>
         <zipCode>28070</zipCode>
         <plus4>1234</plus4>
      </address>

   </distributionCenter>
</getDistributionCenterResponse>
```

3. Product Search Service

3.1 Description:

The Product Search Service performs two functions:

(1) Allows the dealer to view available product brands

(2) Retrieves available products based on user-specified search criteria

3.2 Functional Considerations

3.2.1 What is an ATD Product?

An ATD product is composed of a number of attributes categorized by functional part. Figure 1 (below) illustrates these functional parts and associated attributes of each.

Table 11: ATD Product Parts describes the high-level functionality of each of these parts.

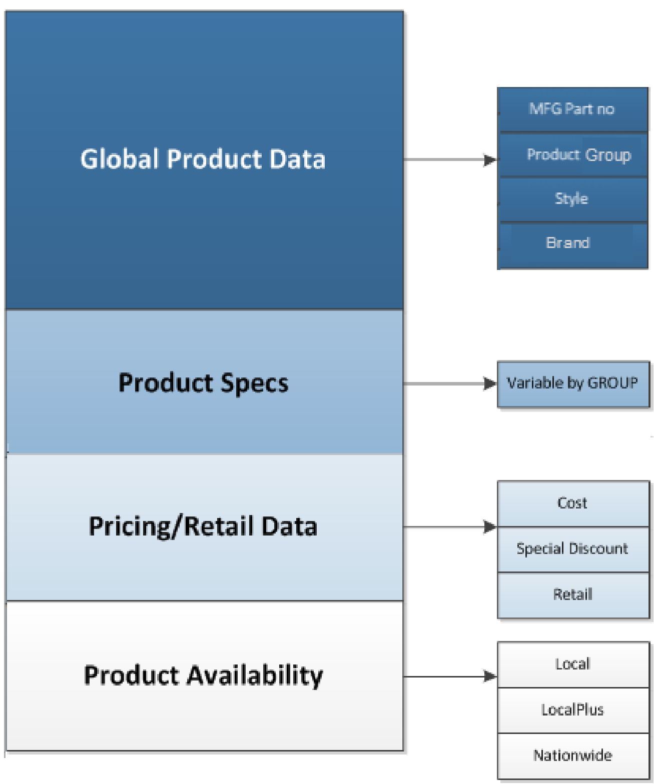

Figure 1: ATD Product parts

Table 11: ATD Product Parts

ATD Product		
Product Part Area	**Product Part Subcomponent**	**Description**
Global Product Data	MFG Part Number	Manufacturer supplied part number. This is generally the part number that is on the product label.
	Product Group	One of 11 ATD-specified product groups. See Table 15 for product group names.
	ATD Product Number	This is a unique, ATD specific part number that maps to Manufacturer's part numbers.
	Style	Manufacturer specified product style name, i.e., Potenza
	Brand	Product brand. i.e. Bridgestone
Product Specs	*Varies by product group*	See Product Search Request Keys for a full list of allowable product specs
Pricing / Retail Data	Cost	Wholesale, dealer price
	Discounts	Vendor-specific discounts
	Retail	Retail price as configured in ATDOnline for the selected location
Availability		The quantity of inventory for products within each regional area (Local, LocalPlus, Nationwide)

3.2.2 Product Images

ATDConnect has the ability to return product image URLs. Images are available in the following 3 sizes:

- Thumbnail – 50x50 pixels
- Small – 138x138 pixels
- Large – 250x250 pixels

When the user requests product image URLs, they will be returned for all product results in the response.

See the Product Images Sample XML Payload for using this feature.

3.3 WSDL Location

Production:	https://ws.atdconnect.com/ws/3_2/products.wsdl
Sandbox/Test:	https://testws.atdconnect.com/ws/3_2/products.wsdl

3.4 Service Method Calls

3.4.1 Get Brands (getBrand)

3.4.2 Function Name

getBrand

3.4.3 Description:

Function for retrieving product brands that are available to the Dealer.

Brands can be filtered by product group by specifying a product-group name in the request.

3.4.4 Request Parameters

Table 12: getBrand request parameters

Parameter Name	Type	Required	Sample / Format	Description
locationNumber	xs:string	yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location

productGroup	xs:string	no	Medium Truck Tire	Used for filtering results by Product Group. See Table 16 valid product group names. Leaving this field blank will result in Brands for all product groups being returned. **Note**: This field is case sensitive

See the Sample XML Payload for creating this request

3.4.5 Request Keys

No name-value pairs for this function exist

3.4.6 Response Parameters

See the Sample XML Payload Response on page 21 for xml formatting:

Table 13: getBrand response parameters

Parameter Name	Type	Sample / Format	Description
XML BrandGroup type			
productGroup	xs:string	Passenger Tires	See Table 16 for valid product group names **Note**: This field is case sensitive
Brands			Collection of brand types
Brand	Brand Type		
XML Brand Type			
Brand	xs:string	Dunlop	Brands available to Dealer location. Not all brands are

			available to every Dealer location

3.4.7 Error Messages

Table 14: getBrand error messages

Error Condition	Error Message
Invalid location	Please enter a valid location
No location entered	Please enter a location
No Brands found	No brands were found matching the entered Criteria

3.4.8 Sample XML Payload

(1) Get Brands w/Product Group Filter

- **Request**


```
<getBrandRequest>
    <brandRequest>

          <locationNumber>434471</locationNumber>
          <productGroup>Passenger Tire</productGroup>

     </brandRequest>
</getBrandRequest>
```


- **Response**


```
<getBrandResponse>

   <brandGroups>
     <brandGroup>

       <productGroup>Passenger Tire</productGroup>
       <brands>
```

```
        <brand>Goodyear</brand>
        <brand>BFGoodrich</brand>
        ...
      </brands>

    </brandGroup>
  <brandGroups>
</getBrandResponse>
```

3.4.9 Search Products by Criteria (getProductByCriteria)

3.4.10 Function Name

getProductByCriteria

3.4.11 Description:

This service searches the ATD platform for available products based on specified search criteria.

Searches may be performed on a multitude of product specification attributes and ATD Global Product data.

Response structure and data can be tailored to fit the user's needs by setting specific option parameters in the request.

Please read the Functional Considerations section below to understand nuances of this function's operation.

3.4.12 Functional Considerations

There are a number of **important** considerations that need to be addressed to implement the Product Search function correctly.

General Product Search Considerations

Guidelines that need to be followed for this function to operate correctly:

- Search criteria are specified using the 'criteria' parameter with corresponding key-value pairs.

- Criteria keys that represent tire products cannot be used to search for wheel products (See the Product Group subsection on page 25). For instance, searching the Product Group ‘Passenger Tires’ with a Wheel group key such as ‘WheelDiameter’ will throw an exception.
- Some search criteria allow you to specify **wildcard (*)** characters in the value field.
    - ATDProductNumber values accept wildcards (*) at the end and/or beginning of the value string. A **minimum of 3 alphanumeric** characters are required.
    - MFGProductNumber values accept wildcards (*) at the end and/or beginning of the value string. A **minimum of 3 alphanumeric** characters are required.
    - [Tire] Size values accept wildcards (*) at the **end the value string**.
- The response payload structure can be controlled by specifying desired response elements under options. For instance, ‘price’ is not a default output attribute in product search, so to view all price options in the response, include the ‘<price/>’ tag inside the ‘<options/> request.
- Product Number criteria keys (ATDProductNumber, MFGProductNumber) can be specified multiple times, but different product number types cannot be specified in the same request. For example:

```
<criteria>
   <entry>
       <key>ATDProductNumber</key>
       <value>1234</value>
   </entry>
   <entry>
       <key>ATDProductNumber</key>  <= OK
       <value>24567</value>
    </entry>
    <entry>
       <key>MFGProductNumber</key> <= NOT OK!
       <value>8901</value>
   </entry>
</criteria>
```

3.4.12.1 Required Search Fields

To limit the number of search results, there are required combinations of request fields to be used in the request.

Table 15 lists the different combinations of required fields needed to search.

Table 15: Product Search Required Fields

Request Field/Key	Request Field/Key	Request Field/Key	Description
Criteria / ATDProductNumber *OR* criteria / ATDLegacyProductNumber (deprecated) *OR* criteria / MFGProductNumber + criteria / Brand	*none*	*none*	Searches can be made on specific product numbers. When using MFGProductNumber, make sure to also use Brand as an additional criteria. This is not required but is highly advised due to manufacturers using the same product number across brands.
Criteria / Size			Tire products may be searched using only the tire size.
Criteria / ProductGroup	Criteria / Brand		All products belonging to a particular brand may be retrieved using both the *brand* and *ProductGroup* field. For example, "Passenger Tires" and "Goodyear"
Criteria / ProductGroup	Criteria / WheelDiameter		Wheel products may be searched using the *ProductGroup* field and the *wheel diameter*.
Criteria / ProductGroup	Criteria / BoltPattern		Wheel products may be searched with the wheel's bolt pattern.
Criteria / ProductGroup	Criteria / NumberOfLugHoles	Criteria / BoltCircle	The wheel's bolt pattern can also be determined from the *Number of Lug Holes* and *Bolt Circle* fields.

3.4.12.2 Product Groups

ATD products fall into one of the 11 ATD *Product Groups* listed below. These groups can be classified as belonging to either 'Tire' or 'Wheel' categories. Each category has its own set of product attribute keys listed in Table 17 and Table 18 below.

Attributes from different Product categories cannot be used in the same search. For instance, 'WheelDiameter' cannot be used with 'WinterDesignation' from the Tire category.

Not all attributes will always be available for every product. ATD will return the attributes we have data available for.

Table 16: ATD Product Groups (colors match keys in Table 22: getProductByCriteria Criteria Keys - Product Specs)

Product Spec Categories	Product Group	Example Criteria Keys
Tire Specs	Passenger Tire	• AspectRatio • Asymmetrical • Temperature Rating
	Light Truck Tire	
	Medium Truck Tire	
	Small Tire	
	Boat Trailer Tire	
	Farm Tire	
	Industrial Tire	
	OTR, Grader, EM Tires	
Wheel Specs	Ag Wheels	• Lug • WheelDiameter • HubBore
	Steel Wheels	
	Custom Wheels	

Table 17: Criteria ProductSpec Request Keys

Request Product Spec Categories			
Tire Spec Keys		Wheel Spec Keys	
AspectRatio	Sidewall	ATDFinish	Material
AssemblyFlag	SingleMaxLoadLbs	BackSpacing	MFGFinish
Asymmetrical	Size	BoltCircle	NumberOfLugHoles

Directional	SmartwayDesignation	BoltPattern	Offset
DualMaxLoadLbs	SpecialNotes	HubBore	OffsetDescription
LoadIndex	SpeedRating	HubCentricFlag	ReplacementCapPart Number
MaxRimWidth	Style	HubCentricRingIncud edFlag	SpecialNotes
MaxSpeed	TemperatureRating *see note below	HubCover	Style
MaxTirePressure	TireWidth	LoadRating	WheelDiameter
MileageWarranty	TractionRating *see note below	LugNutType	WheelWidth
MinRimWidth	TreadDepth		
OuterDiameter	TreadWear *see note below		
RadialBiasFlag	TubelessFlag		
RimDiameter	UTQGL *see note below		
RPM	WinterDesignation		
RunFlat	WinterStudSize		
SetFlapFlag	WinterDesignation		
SetTubesAndFlapFla g			

*TemperatureRating, TractionRating, and TreadWear make up the UTQGL specification. UTQGL is not a valid productSpec key on its own.

Table 18: Criteria ProductSpec Response Keys

Request Product Spec Categories			
Tire Spec Keys		**Wheel Spec Keys**	
AspectRatio	Sidewall	ATDFinish	Material
AssemblyFlag	SingleMaxLoadLbs	BackSpacing	MFGFinish

Asymmetrical	Size	BoltCircle	NumberOfLugHoles
Directional	SmartwayDesignation	BoltPattern	Offset
DualMaxLoadLbs	SpecialNotes	HubBore	OffsetDescription
LoadIndex	SpeedRating	HubCentricFlag	ReplacementCapPart Number
MaxRimWidth	Style	HubCentricRingIncud edFlag	SpecialNotes
MaxSpeed	TemperatureRating *see note below	HubCover	Style
MaxTirePressure	TireWidth	LoadRating	WheelDiameter
MileageWarranty	TractionRating *see note below	LugNutType	WheelWidth
MinRimWidth	TreadDepth		
OuterDiameter	TreadWear *see note below		
RadialBiasFlag	TubelessFlag		
RimDiameter	UTQGL *see note below		
RPM	WinterDesignation		
RunFlat	WinterStudSize		
SetFlapFlag	WinterDesignation		
SetTubesAndFlapFla g			

*TemperatureRating, TractionRating, and TreadWear make up the UTQGL specification. UTQGL is not a valid productSpec key on its own.

3.4.13 Request Parameters

Table 19: getProductCriteria request parameters

Parameter Name	Type	Required	Sample / Format	Description

locationNumber	xs:string	yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location
criteria	Criteria Type	no	See Table 20 for valid key options	Specify only 1 key type in Table 21 to search products using the key's value. These keys are the ATD Global Product data types
options	Options Type	no	Include empty element tags to include option. <availability/> <price/>	Specifying options parameters returns only the specified option in the response. For example: <options><price/><options> Will return only the price data in the response.
XML Options Type				
price	Price Type	no	An empty tag <price/>	Specifies all prices to be included in response
availability	Inventory Type	no	An empty tag <availability/>	Specifies all quantities of available products with different delivery options to be included in response
images	Images Type	no	<images> 1 </images>	The example specifies that only URLs for small images will be returned. An empty <images/> node will result in no image URLs being returned.
productSpec	Criteria Type	no	<productSpec> <entry> <key>AspectRatio</key> </entry> </productSpec>	The example specifies that the response structure will only include the key-value field 'Aspect Ratio'. An empty <productSpec/> node will return all available keys for the product.
XML Price Type				

price	xs:decimal	no	<price>1</price>	Specifies only price to be included in response
retail	xs:decimal	no	<retail>1</retail>	Specifies only retail price to be included in results
specialDiscount	xs:decimal	no	<specialDiscount>1</specialDiscount	Specifies only special discount price to be included in results
XML Inventory Type				
local	xs:int	no	<local>1</local>	Specifies only quantities of product at local distribution center to be included in response
localPlus	xs:int	no	<localPlus>1</localPlus>	Specifies only quantities of product at select adjacent distribution centers to be included in response
nationWide	xs:int	no	<nationwide>1</nationwide>	Specifies only quantities of product nationwide to be included in response
XML Images Type				
thumbnail	xs:int	no	<thumbnail>1</thumbnail>	Specifies thumbnail-sized product image URLs will be returned See Product Images
small	xs:int	no	1	Specifies small-sized product URLs will be returned See Product Images
large	xs:int	no	<large>1</large>	Specifies large-sized product image URLs will be returned See Product Images

See the Sample XML Payload for creating this request

3.4.14 Request Keys

The product search request keys are all specified inside the 'criteria' element. However, for clarity, they are broken down here into separate tables by function:

- Table 20 Product Number Keys
- Table 21 Global Product Data Keys
- Table 22 ProductSpec Keys

Table 20: getProductByCriteria Criteria Keys - ProductNumbers

Key	Value Type	Required	Example	Description
ATDProductNumber	xs:string	No	702524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
MFGProductNumber	xs:string	No	ABC9384	Manufacturer supplied part number. This is generally the part number that is on the product label. It is not unique across all items. It can contain special characters.
ATDLegacyProductNumber	xs:string	No	012345678	This is the unique, 9-digit ATD legacy part number. ** This number is deprecated and will be removed from future versions. ** It is supported in ATDConnect requests, but responses do not return the parameter.

Table 21: getProductByCriteria Criteria Keys - Global Product Data

Key	Value Type	Required	Example	Description
Brand	String	no	Goodyear	Use the Get Brands function to view an available list of brands
ProductGroup	String	no	Passenger Tires	See Table 16 for valid Product group names **Note**: This field is case sensitive
Availability	String	no	Local	Specifying the values: • Local • LocalPlus Returns only products available at local or regional distribution centers. Within area, 'out of stock' products are not returned.

Table 22: getProductByCriteria Criteria Keys - Product Specs

Product Group	Key	Value Type	Example	Description
	Size	xs:string	205/60R15	The tire size. Values are stored in the format : 205/60R15 Where: 205 is the section width in mm 60 is the aspect ratio R is the type of construction (Radial) 15 is the rim diameter in inches
	BoltCircle	xs:decimal	4.50 or 120	The bolt circle is the diameter of a circle formed by the centers of the lugs on your automobile's wheel.

Product Group	Key	Value Type	Example	Description
				The diameter can be specified in either imperial or metric values. This is a numeric value only.
	BoltPattern	xs:string	4 x 112	Bolt pattern of wheel. See definition for 'Available Bolt Patterns'
	NumberOfLugHoles	xs:int	5	This value corresponds to the number of lug holes on a wheel.
	WheelDiameter	xs:decimal	18	Diameter of wheel specified in inches

3.4.15 Response Parameters

See the Sample XML Payload Response for xml formatting

Table 23: getProductCriteria response parameters

Parameter Name	Type	Sample / Format	Description
products	Parent type for Product Type collection		Collection of products returned in search
XML Product Type			
ATDProductNumber	xs:string	702524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
MFGProductNumber	xs:string	ABC9384	Manufacturer supplied part number. This is generally the part number that is on the product label. It is not unique across all items. It can contain special characters.

style	xs:string	EXPLORER PLUS	Manufacturer specified product style name
brand	xs:string	Dunlop	Product brand. i.e. Bridgestone
productGroup	xs:string	Light Trucks	ATD product type classification of product. See Table 16 for valid Product group names **Note**: This field is case sensitive
productSpec	Criteria Key/Value type		Key value pairs of detailed product specifications. See Table 22 for value values.
price	Price type		Pricing details of product
availability	Inventory Type		Availability of product at area Distribution Centers
images	Images Type		Product image URLs
XML Price Type			
price	xs:decimal	54.46	Wholesale price to the customer.
retail	xs:decimal	69.46	Suggested retail price.
specialDiscount	xs:decimal	0.07	Pricing discount as a percentage; for example, 0.07 = 7% off the price
XML Inventory Type			
local	xs:int	1	Quantity of product at local distribution center
localPlus	xs:int	3	Quantity of product at selected adjacent distribution centers
nationwide	xs:int	100	Quantity of product at all distribution centers nationwide

XML Images Type			
thumbnail	xs:anyURI	<thumbnail>http://www.atdonline.com/medias/celum_assets/8824522309662_MichelinDefender_977_977_jpg.jpg?131</thumbnail>	The URL of the thumbnail-sized product image See Product Images
small	xs:anyURI	http://www.atdonline.com/medias/celum_assets/8824521293854_MichelinDefender_977_977_jpg.jpg?132	The URL of the small-sized product image See Product Images
large	xs:anyURI	<large>http://www.atdonline.com/medias/celum_assets/8824520179742_MichelinDefender_977_977_jpg.jpg?134</large>	The URL of the large-sized product image See Product Images

3.4.16 Response Keys

In addition to all the keys specified in the request (above), the following Product Spec keys may appear in the response.

Table 24: getProductCriteria response keys

Product Group	Key	Value Type	Example	Description
Tires	AspectRatio	xs:decimal	55	Ratio of sidewall height to tire width.
	AssemblyFlag	xs:string	N	A combined tire and wheel assembly. Primarily used for trailer tires. Allowable values: • Y • N
	Asymmetrical	xs:string	Y	Tire must be mounted with a specific side on the outside of the car.

Product Group	Key	Value Type	Example	Description
				Allowed values: • Y
	Directional	xs:string	Y	Tires designed to rotate in only one direction Allowed values: • Y • N
	DualMaxLoadLBS	xs:decimal	8,270	Tire load amount when tire is used in dual assemblies. Specified in lbs.
	MaxRimWidth	xs:decimal	7.5	Maximum size of rim width for tire fitment specified in inches This is the maximum distance between the two opposite inside edges of the rim flanges that will allow the tire to be safely mounted.
	MaxSpeed	xs:int	75	Maximum speed a tire can carry its load specified in miles per hour
	MaxTirePressure	xs:decimal	33	Maximum tire pressure in PSI
	MileageWarranty			
	MinRimWidth	xs:decimal	6.5	Minimum size of rim for tire fitment specified in inches This is the minimum distance between the two opposite inside edges of the rim flanges that will allow the tire to be safely mounted.
	OuterDiameter	xs:decimal	40.5	The diameter of an inflated tire without any load. Specified in inches
	RadialBiasFlag	xs:string	R	Whether the tire construction is Radial or Bias. Allowable values: • R • B

Product Group	Key	Value Type	Example	Description
	RimDiameter	xs:int	16	The diameter of the rim bead seats supporting the given tire. Value specified in inches.
	RPM	xs:int	774	Tire revolutions per mile
	Runflat	xs:string	Y	A run-flat tire allows the car to be driven for limited distances without any air in the tire Allowed values: • Y • N
	SetFlapFlag	xs:string	Y	Flaps protect the tube from chafing on the rim. Designed for flat-based rims. This field designates if a flap is included with the tube. Allowable Values: • Y • N
	SetTubesAndFlapFlag	xs:string	Y	This field designates if the tire comes with a flap and/or tube
	Sidewall	xs:string	Blackwall	Type of sidewall. Allowable values: See sidewall chart in Appendix 2 on page 93
	SingleMaxLoadLBS	xs:string	2601	Maximum tire load of a single tire assembly specified in LBS
	Size	xs:string	205/60R15	The tire size. Values are stored in the format : 205/60R15 Where: 205 is the section width in mm 60 is the aspect ratio

Product Group	Key	Value Type	Example	Description
				R is the type of construction (Radial) 15 is the rim diameter in inches
	SpecialNotes	xs:string	NHS - Non Highway Service	Special notes about the product. These can be many values depending on the type of item.
	SpeedRating	xs:int	H	The speed category or range of speeds a tire can carry load under specified conditions. Allowable values: See Speed Rating Chart in Appendix 2 on page 90
	Style			
	SmartwayDesignation	xs:string	Y	SmartWay recognizes products that reduce vehicle-related emissions. Smartway tire models can reduce emissions and improve fuel efficiency. Allowable values: • Y • N
	TemperatureRating	xs:string	A	A tire's resistance to heat. Allowable values: • A • B • C
	TirePly	xs:string	C	Plies are the different layers of belting that make up the tire body. Allowable values: See Tire Ply Chart in Appendix 2 on page 90

Product Group	Key	Value Type	Example	Description
	TireWidth	xs:int	225	Width of the tire in millimeters.
	TractionRating	xs:string	AA	Tire's ability to stop on wet pavement. Allowable values: • AA • A • B • C
	TreadDepth	xs:int	13	Depth of tire tread specified in 1/32 of an inch
	TreadWear			
	TubelessFlag	xs:string	Y	Tires that do not require a separate inner tube. Allowable values: • Y • N
	UTQGL	xs:string	260AA	Uniform Tire Quality Grade Labeling (UTQGL) is a system that classifies tires as to tread wear, traction and heat resistance.
	WinterDesignation	xs:string	Winter	Specifies if tire is designed for Winter driving Allowable values: • Winter • 3 Peak Mt. Snowflake • All Other
	WinterStudSize	xs:string	250 × 185	Stud sizing in mm.
	ATDFinish	xs:string	Chrome	ATD specified wheel finishes. These are generically labeled finishes such as 'Chrome' or 'Black' that are more generic than the detailed manufacturer finishes

Product Group	Key	Value Type	Example	Description
				Allowable values: See ATD Finish Chart in Appendix 2 on page 92
	BackSpacing	xs:decimal	5.25	Distance from the hub mounting surface to the centerline of the wheel. Specified in inches.
	BoltCircle	xs:decimal	4.50 or 120	The bolt circle is the diameter of a circle formed by the centers of the lugs on your automobile's wheel. The diameter can be specified in either imperial or metric values. This is a numeric value only.
	BoltPattern	xs:string	4 x 112	Bolt pattern of wheel. See definition for 'Available Bolt Patterns'
	HubBore	xs:decimal	67.1	The centerbore of the wheel, hub diameter, or size of the hole in the back of the wheel that centers it over the mounting hub. Specified in mm.
	HubcentricFlag	xs:string	Y	Hubcentric wheels have a centerbore that matches the hub exactly. Allowable values: • Y • N
	HubcentricRingIncludedFlag	xs:string	Y	The hub centric ring fills the gap between the hub of the car and center bore of the wheel. Many aftermarket custom wheels require hub centric ring to prevent off-centering. This value indicates if a hub centric ring is included with the wheel

Product Group	Key	Value Type	Example	Description
				Allowable values: • Y • N
	HubCover	xs:string	Y	Decorative cover that hides the vehicle hub. This value indicates whether the wheel comes with a hub cover Allowable values: • Y • N
	LoadRating	xs:int	750	The amount of weight the wheel can sustain. Specified in lbs.
	LugNutType	xs:string	Cone seat	The style of the lug nut or bolt that hold the wheel onto the car.
	Material	xs:string	Aluminum	This is the material composition of the wheel. These are typically aluminum
	MFGFinish	xs:string	Black	Manufacturer specified wheel finish. There are many independent values associated with this field, however each field maps to a generic, ATD-specified 'ATD Finish' Some examples of MFG Finish are: • Anthracite Grey w/Diamond Cut and Clear Coat • Bead Blasted Natural Aluminum with Machined Accents Black with Red Racing Stripe
	NumberOfLugHoles	xs:int	5	This value corresponds to the number of lug holes on a wheel.

Product Group	Key	Value Type	Example	Description
	Offset	xs:decimal	35.00	Distance between the hub mounting surface and the center line of the rim. Specified in mm
	OffsetDescription	xs:string	MED	Descriptive grouping of wheel offsets.
	ReplacementCapPartNumber	xs:string	CFW10101675	Hub caps or wheel cover product number. Specified as an alphanumeric string
	Special Notes	xs:string		Special notes about the product. These can be many values depending on the type of item
	Style	xs:string		
	WheelDiameter	xs:decimal	18	Diameter of wheel specified in inches
	WheelWidth	xs:decimal	5.5	Distance between inside of flanges rounded to nearest 1/2". Specified in inches.

3.4.17 Error Messages

Table 25: getProductByCriteria error messages

Error Condition	Error Message
Invalid criteria (ATDProductNumber, MFGProductNumber, etc.)	Invalid criteria specified. Please correct your request.
Invalid product specs	Invalid product specification requested. Please correct your request.
Missing request key pair. E.g. bolt circle and number of lug holes	Invalid criteria specified for a tire request. Size or brand is required

Invalid location number	Please enter a valid location
Empty location	Please enter a location

3.4.18 Sample XML Payload

(1) Product Search with *Global Product* criteria

This example will search the product catalog looking for products where:

ATDProductNumber = 12345 or ATDProductNumber like '2345 (wildcard)

Note: Searches using product numbers will ignore all other criteria such as 'Brand' and 'ProductGroup'.

- **Request**


```
<getProductByCriteriaRequest>
  <locationNumber>434471</locationNumber> <=Required field
  <criteria>

    <entry>
      <key>ATDProductNumber</key>
      <value>12345</value>
    </entry>

    <entry>
      <key>ATDProductNumber</key> <= same key type OK. Performs 'OR' operation
      <value>2345*</value> <= wildcard search
    </entry>

    <!-- invalid entry -->
    <!--
    <entry>
      <key>MFGProductNumber</key> <= Cannot use multiple criteria key types
      <value>12345</value>
    </entry>
    -->

  </criteria>
</getProductByCriteriaRequest>
```

- **Response**

```
<getProductByCriteriaResponse>

   <products>
     <product>
       <ATDProductNumber>12345</ATDProductNumber>

       ... other Global product elements

       <brand>Dunlop</brand>
                <productGroup>Passenger Tire</productGroup>

       ... other product elements
     </product>

     <product>

       <ATDProductNumber>23456789</ATDProductNumber> <= wildcard
                                   match

       ... other Global product elements

       <brand>Dunlop</brand>
                <productGroup>Passenger Tire</productGroup>
       ... other product elements
     </product>

   </products>

</getProductByCriteriaResponse>
```

(2) <u>Product Search with options</u>

This request will use the *Options* tag to structure the elements in the response

- **Request**

```
<getProductByCriteriaRequest>
  <locationNumber>434471</locationNumber> <=Required field
  <criteria>
        <entry>
                <key>Brand</key>
                <value>Goodyear</value>
        </entry>
```

```
          <entry>
                    <key>ProductGroup</key>
                    <value>Passenger Tire</value>
          </entry>
   </criteria>

   <options>

     <price/> <= include all price options in response

     <availability>

        <local>1</local> <= include only local availability in response.  Omit 'localPlus' and 'nationWide'
     </availability>

   </options>
</getProductByCriteriaRequest>
```

- **Response**

```
<getProductByCriteriaResponse>

   <products>
      <product>
        <ATDProductNumber>12345</ATDProductNumber>

        ... other Global product elements

        <brand>Goodyear</brand>
                    <productGroup>Passenger Tire</productGroup>

        <productSpec>
            ...product specifications
        </productSpec>

        <price>   <= return all price elements
           <price>30.00</price>
           <retail></retail>
           <specialDiscount></specialDiscount>
        </price>

        <availability> <=return only 'local' inventory elements
           <local>12</local>
        </availability>

      </product>
   </products>
```

```
</getProductByCriteriaResponse>
```

(3) <u>Product Search different *Product Groups* Error</u>

This request will result in an error because *productSpec* keys from different product groups are used.

- **Request**

```
<getProductByCriteriaRequest>

  <locationNumber>434471</locationNumber> <=Required field

  <criteria>
   ...
        <entry>
     <key>Brand</key>
     <value>Dunlop</value>
   </entry>

        <entry>
     <key>ProductGroup</key>
     <value>Passenger Tire</value>
   </entry>

        <entry>
     <key>MaxTirePressure</key>  <= Key from Tire group
     <value>33</value>
   </entry>

   <entry>
     <key>NumberOfLugHoles</key>  <= Key from wheel group. Incompatible
     <value>5</value>
   </entry>

  </criteria>

  <options>
   <availability>
      <local>1</local>
   </availability>
  </options>

</getProductByCriteriaRequest>
```

- **Response**

```
<soap:Envelope    xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Header/>
  <soap:Body>
   <faultcode>soap:Server</faultcode>
   <faultstring>ATDConnectException</faultstring>
   <detail>
     <message>
       Invalid product criteria request. You cannot mix <= ERROR
       Tire and wheel criteria
     </message>
   </detail>
  </soap:Body>
</soap:Envelope>
```

(4) Product Search - return tire products available at the local distribution center

- **Request**

```
<getProductByCriteriaRequest>

  <locationNumber>434471</locationNumber> <=Required field

  <criteria>
   ...
   <entry>
     <key>Brand</key>
     <value>Dunlop</value>
   </entry>

   <entry>
     <key>ProductGroup</key>
     <value>Passenger Tire</value>
   </entry>

   <entry>
     <key>Size</key>  <= Key from Tire group
     <value>P225*</value>  <= Wildcard value
   </entry>

   <entry>
```

```
        <key>Availability</key>  <= Key from wheel group. Incompatible
        <value>Local</value>
      </entry>

    </criteria>

    <options>
     <availability>
         <local>1</local>
      </availability>
    </options>

</getProductByCriteriaRequest>
```


- **Response**


```
<getProductByCriteriaResponse>
    <products>
        <product>
          <ATDProductNumber>142000487</ATDProductNumber>
          <MFGProductNumber>356142443</MFGProductNumber>
          <productGroup>Passenger Tires</productGroup>
          <style>EXPLORER PLUS</style>
          <brand>Dunlop</brand>
          <availability>
             <local>2</local>
          </availability>
        </product>
        <product>
          <ATDProductNumber>096000052</ATDProductNumber>
          <MFGProductNumber>WN630</MFGProductNumber>
          <productGroup>Passenger Tires</productGroup>
          <style>NEG S-1023 - 409</style>
          <brand>Dunlop</brand>
             <availability>
             <local>23</local>
          </availability>
        </product>

         ... many more products

    </products>
</getProductByCriteriaResponse>
```


(5) Product Search - requesting product image URLs

- **Request**

```
<getProductByCriteriaRequest>
        <locationNumber>434471</locationNumber>
        <criteria>
          <entry>
            <key>MFGProductNumber</key>
            <value>86105</value>
          </entry>
        </criteria>
        <options>
          <images>
            <thumbnail>0</thumbnail>
            <small>1</small>
            <large>0</large>
          </images>
        </options>
</getProductByCriteriaRequest>
```

Use "1" to return image URLs for the desired size. Using "0" or leaving blank will result in no image URLs being returned.

- **Response**

```
<getProductByCriteriaResponse>
        <products>
          <product>
            <ATDProductNumber>86105</ATDProductNumber>
            …
            <images>
              <thumbnail/>
              <small>
                <images id="1">http://www.atdonline.com/medias/celum_assets/8824031281182_
                  michelin_defender_bsw_asymmetrical_386x386_1275_jpg.jpg?129</images>
                <images id="2">http://www.atdonline.com/medias/celum_assets/8824191909918_
                  michelin_defender_bsw_asymmetrical_detail_386x386_697_jpg.jpg?132</images>
                <images id="3">http://www.atdonline.com/medias/celum_assets/8824358043678_
                  michelin_defender_bsw_asymmetrical_sidewall_386x386_955_jpg.jpg?131</images>
                <images id="4">http://www.atdonline.com/medias/celum_assets/8824521293854_
                  MichelinDefender_977_977_jpg.jpg?132</images>
                <images id="5">http://www.atdonline.com/medias/celum_assets/8828210053150_
                  MichelinDefender_2817_2817_jpg.jpg?132</images>
              </small>
              <large/>
            </images>
          </product>
        </products>
</getProductByCriteriaResponse>
```

Return to [Functional Considerations: Product Images]

4. Order Service

4.1 Description:

The ATDConnect Order service provides mechanisms for previewing and placing ATD orders.

It consists of two functions: previewOrder and placeOrder.

The previewOrder function is an intermediary step that can be used for verifying and confirming order contents and availability.

The placeOrder function finalizes the order submission process by sending the order details to ATD for processing.

Neither function has any dependencies on values from the other.

4.2 Functional Considerations

4.2.1 'Fill and Kill' / 'Fill or Kill'

The *FillKill* option in order services specifies how an order with unavailable items should be fulfilled.

fillKill can have 2 values:

- fok = fill or kill
- fak = fill and kill

This parameter may be set at either the *Order level* or at the *Order Line level.*

At the ***Order Level:***

- fok: if any item in the order is unavailable (status = backordered), the entire order is cancelled; otherwise the order is filled
- fak: unavailable items (status = backordered) are removed from the order. Available orders, including those with partial quantities are processed.

At the ***Order Line Level:***

- fok: if any line item order is unavailable (status = backordered) the entire line item is removed from the order, even if partial quantities can be filled.
- fak: all available items at the line level are processed, including partial quantities

If *fillKill* is null, "backordered" products are processed by closest LocalPlus or Nationwide distribution centers.

This functionality is illustrated by the cases below:

<u>Case 1:</u>

An order is placed for 2 items, productA and productB.

The user specifies FOK at the order level and requests 10 items of productA and 2 items of productB.

At the local distribution center there are 20 items of productA available, but only 1 item of productB.

The order request for productB cannot be filled because there is not enough quantity in stock.

Since FOK (fill or kill) has been specified at the order level, the whole order is killed.

Result: No order is placed.

Case 2:

An order is placed for 2 items, productA and productB.

The user requests 10 items of productA and 2 items of productB. The user specifies FAK (fill and kill) at the order-line level for productB.

At the local distribution center there are 20 items of productA available, but only 1 item of productB.

The order request for productB cannot be completely filled because there is not enough quantity in stock.

Since FAK has been specified for productB, the 1 available item of productB will be filled with the order along with the 10 items of productA.

Result: Partially filled order – 10 productA and 1 productB (1 productB item killed).

See the #2 and #3 Sample XML Payload to view this functionality

4.2.2 Quick Ship

The *quickShip* parameter expands the product availability search beyond the Local and Regional Distribution Centers to National locations.

Additional charges are incurred when a product is ordered from Nationwide Distribution centers, and typical delivery times range from 2 to 3 days.

When *quickShip* is used in conjunction with the *fillKill* parameter (see prior section), items and orders are only killed if they are unavailable at both the Regional and National Distribution Center locations.

quickShip can have 2 values:

- Y
- N

The *quickShip* parameter will default to N if it is left blank or omitted entirely from the request. A value of "N" will result in quickShip not being allowed.

This functionality is illustrated by the cases below:

> **<u>Case 1:</u>**
>
> As with the FOK illustration above, an order is placed for 2 items, productA and productB.
>
> The user specifies FOK at the order level and requests 10 items of productA and 2 items of productB. The user also specifies ‘Yes’ to the quickShip parameter.
>
> At the local distribution center there are 20 items of productA available, but only 1 item of productB. However nationally, there are 400 items of productB available.
>
> The order request for productB cannot be filled locally because there is not enough quantity in stock. Quickship tell the order fulfillment process to retrieve an extra item of productB from the closest distribution center with available product.
>
> Although FOK (fill or kill) has been specified, the order can be filled because the user has also specified QuickShip. Please note that the single item of productB sourced from an alternate Distribuition center may mean the order incurs freight charges.
>
> **Result:** Order is filled with freight costs for 1 item.

4.2.3 LocalPlus

The *localPlus* parameter enables the user to allow or deny orders being filled from LocalPlus inventory. Products filled from LocalPlus inventory ship at no cost to the user but they can often have slightly longer delivery times than products filled from Local inventory.

localPlus can have 2 values:

- Y
- N

The user can use this parameter in conjunction with the *quickShip* parameter (see prior section) to control exactly where their orders are allowed to be filled from (Local, LocalPlus, National). **Use the following combinations to achieve the desired result:**

- For Local inventory only
  - *localPlus* = N
  - *quickShip* = N **OR** omit *quickShip*
- For Local and LocalPlus inventory only
  - *localPlus* = Y **OR** omit *localPlus*
  - *quickShip* = N **OR** omit *quickShip*
- For Local, LocalPlus, and National inventory
  - localPlus = Y **OR** omit *localPlus*

- *quickShip* = Y

Note: when omitted, localPlus defaults to Y and quickShip defaults to N.

4.2.4 Cart Line Number

The cart line number is used in order to assign a unique value to each product in the customer's cart. There are 2 ways that the cart line number is generated:

1) ATD will automatically generate one for each product in the order they appear in the cart. The cart line number will begin at 1 and increase consecutively.
2) The user can assign the cart line numbers themselves using the cartLineNumber parameter.

The cart line number can be assigned in the previewOrder and placeOrder calls and will be returned in their responses. It will be also be returned along with order details when using getOrderDetail to search for orders that have already been placed.

Note: if the user does not provide a specific cartLineNumber in the previewOrder call, a cartLineNumber of 0 will be returned for each product in the cart.

4.3 WSDL Location

Production:	https://ws.atdconnect.com/ws/3_2/orders.wsdl
Sandbox/Test:	https://testws.atdconnect.com/ws/3_2/orders.wsdl

4.4 Service Method Calls

4.4.1 Preview an Order (previewOrder)

4.4.2 Function Name

previewOrder

4.4.3 Description:

Intermediary stage for reviewing an order before it is placed.

This method will validate the submitted order and return an order response with calculated order totals, tax, and delivery data if successful.

This function is not a prerequisite for calling the placeOrder method, and may be called multiple times.

4.4.4 Request Parameters

Table 26: previewOrder request parameters

Parameter Name	Type	Required	Sample / Format	Description
locationNumber	xs:string	Yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location
Order	Order Type	Yes		
XML Order Type				
customerPONumber	xs:string	No (Some clients require this parameter)	69686	Customer-specified order number that gets passed back in response.
pickup	xs:string	No	Y	Whether or not the order will be picked up. Allowable values: • Y • N
fillKill	xs:string	No	fok	FOK will kill the order if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally, and remove unavailable quantity from the order.

				'Fill and Kill' or 'Fill or Kill' order. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
orderType	xs:string	No	D	Dealer or Retail order type. Allowable values: • D • R
consumerData	ConsumerData Type	No		Consumer information (name, address, etc.). **NOTE:** Can only be used with prior coordination with ATD.
lineitems	LineItem Type (*)	No		Collection of order – line items
XML ConsumerData Type				
consumerShipTo	ConsumerInfo Type	No		Ship to details **NOTE:** Can only be used with prior coordination with ATD.
consumerBillTo	ConsumerInfo Type	No		Bill to details **NOTE:** Can only be used with prior coordination with ATD.
consumerAttributes	criteriaType	No		These are client-specified values that get returned in the web service response exactly as they were entered in the request. i.e. 'Pass through' attributes. See Table

				27 for consumer attribute keys **NOTE**: Can only be used with prior coordination with ATD.
XML ConsumerInfo Type				
name	xs:string	No		
phoneNumber	xs:string	No		
emailAddress	xs:string	No		
address	Address Type	No		
XML Address Type				
address1	xs:string	No	123 Main Street	Address line 1
address2	xs:string	No	Suite 1	Address line 2
address3	xs:string	No		Address line 3
city	xs:string	No	Huntersville	
state	xs:string	No	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	No	12345	5-digit U.S. zip code
plus4	xs:string	No	1234	U.S. zip code extension. Ex, 20001-**0001**
XML LineItem Type				
cartLineNumber	xs:int	No	1	This is a number given consecutively to each product in the cart or assigned by the user as assigned. See Cart Line Number

ATDProductNumber	xs:string	Yes	702524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
quantity	xs:int	Yes	1	Quantity of product requested
fillKill	xs:string	No	fok	FOK will kill the order line item if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally, and remove unavailable quantity from the order line item. 'Fill and Kill' or 'Fill or Kill' order line item. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
quickShip	xs:string	No	Y	Quick ship option. See Quick Ship
localPlus	xs:string	No	Y	LocalPlus option. See LocalPlus

4.4.5 Request Keys

Table 27: previewOrder – Consumer Attributes request keys

Key	Value Type	Required	Example	Description

consumerAttribute1	String	No	Any data	Can only be used with prior coordination with ATD.
consumerAttribute2	String	No	More of my data	Can only be used with prior coordination with ATD.
consumerAttribute3	String	no	And more	Can only be used with prior coordination with ATD.

4.4.6 Response Parameters

See the Sample XML Payload Response for xml formatting.

Table 28: previewOrder response parameters

Parameter Name	Type	Sample / Format	Description
order	PreviewOrder Type		
XML PreviewOrder type			
customerPONumber	xs:string	myProdNum	Customer-specified order number that gets returned from request
fillKill	xs:string	fok	FOK will kill the order if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally, and remove unavailable quantity from the order. 'Fill and Kill' or 'Fill or Kill' order line item. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
pickup	xs:string	N	Whether or not the order will be picked up.
orderTotal	xs:string	54.46	Total amount of order.

orderLines	OrderLine type		
consumerData	ConsumerData type		Consumer information (name, address, etc.)
XML OrderLine Type			
cartLineNumber	xs:int	1	This is a number given consecutively to each product in the cart or assigned by the user as assigned. See Cart Line Number
billTo	xs:string	Generic Tire Co.	Name of person or entity to bill the order to
quantity	xs:int	1	Quantity of item requested.
lineTotal	xs:string	54.46	Line item total price in USD
ATDProductNumber	xs:string	382524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
cost	xs:string	54.46	Price of item excluding tax in USD.
fet	xs:string	0.01	Federal Excise Tax amount.
description	xs:string	P215/60R16 Goodride SP06	Product item description
quickShip	xs:string	Y	Quick ship option. See Quick Ship Values: • Y • N
localPlus	xs:string	Y	LocalPlus option. See LocalPlus Values: • Y • N
fulfillments	Fulfillment type		Fulfillment details of a line item. Can include multiple fulfillments

			depending on product availability.
XML Fulfillment Type			
status	xs:string	filled	Availability status for specific quantity of products in line item. Allowable values: • Filled • Killed • Backordered
quantity	xs:int	1	Quantity of items in fulfillment. Quantity <= quantity ordered.
freight	xs:string	25.00	Cost of freight (if any) associated with fulfillment.
shipMethod	xs:string	FEDEX	Shipping method of fulfillment.
estimatedDelivery	xs:dateTime	2013-01-01T08:00:00-5.00GMT	Approximate fulfillment delivery date. With timezone
XML ConsumerData Type			
consumerShipTo	ConsumerInfo Type		Ship to details
consumerBillTo	ConsumerInfo Type		Bill to details
consumerAttributes	criteriaType		**NOTE**: Can only be used with prior coordination with ATD.
XML ConsumerInfo Type			
name	xs:string	John Doe	
phoneNumber	xs:string	123-123-1234	
emailAddress	xs:string	j.doe@domain.com	
address	Address Type		
XML Address Type			
address1	xs:string	123 Main Street	Address line 1

address2	xs:string	Suite 1	Address line 2
address3	xs:string		Address line 3
city	xs:string	Huntersville	
state	xs:string	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	12345	5-digit U.S. zip code
plus4	xs:string	1234	U.S. zip code extension. Ex, 20001-**0001**

4.4.7 Error Messages

Table 29: previewOrder error messages

Error Condition	Error Message
Invalid CustomerPONumber	Invalid PO number used for this request
Invalid location number	Please enter a valid location
Internal Order Error	An internal error occurred. Please contact customer support if this error persists
Invalid ATDProductNumbers used in order	Your request did not return any ATD products
Invalid OrderType value specified	OrderType Unknown, OrderType must be D or R
Invalid FillKill value specified	FillKill Unknown, FillKill must be FOK or FAK
Invalid Pickup value specified	Pickup Unknown, Pickup must be Y or N
Item quantity not specified	Please enter a quantity
Invalid QuickShip value	QuickShip Unknown, QuickShip must be Y or N

Incorrect Quickship usage	Quickship is not available for orders using Pickup
Incorrect FillKill usage	FillKill on Header and Lines, FillKill may not be used on both the header and lines
Invalid FillKill value	FillKill Unknown, FillKill must be FOK or FAK

4.4.8 Sample XML Payload

See the Sample XML Payload examples in the placeOrder function below. These payloads are structured similarly.

4.4.9 Place an Order (placeOrder)

4.4.10 Function Name

placeOrder

4.4.11 Description:

Function for submitting an order to ATD.

If successful, order will return an ATD orderNumber that can be used as a reference for follow up requests.

It is not necessary to call the previewOrder function prior to calling this method.

4.4.12 Request Parameters

Table 30: placeOrder request parameters

Parameter Name	Type	Required	Sample / Format	Description
locationNumber	xs:string	yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location

Order	Cart Type	yes		
XML Cart Type				
customerPONumber		No (Some clients require this parameter)	69686	Customer-specified order number that gets passed back in response.
Pickup		no	N	Whether or not the order will be picked up. Allowable values: • Y • N
fillKill		no	fok	FOK will kill the order if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally, and remove unavailable quantity from the order. 'Fill and Kill' or 'Fill or Kill' order. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
orderType		no	D	Dealer or Retail. Valid values are 'D' or 'R'

consumerData	ConsumerData Type	no		Consumer information (name, address, etc.). **NOTE:** Can only be used with prior coordination with ATD.
Lineitems	LineItem Type (*)	no		Collection of order – line items
XML ConsumerData Type				
consumerShipTo	ConsumerInfo Type	no		Ship to details **NOTE:** Can only be used with prior coordination with ATD.
consumerBillTo	ConsumerInfo Type	no		Bill to details **NOTE:** Can only be used with prior coordination with ATD.
consumerAttributes	criteriaType	no		These are client-specified values that get returned in the web service response exactly as they were entered in the request. i.e. 'Pass through' attributes. See Consumer Attributes **NOTE**: Can only be used with prior coordination with ATD.
XML ConsumerInfo Type				
Name	xs:string	no	John Doe	

phoneNumber	xs:string	no	123-123-1234	
emailAddress	xs:string	no	j.doe@domain.com	
Address	Address Type	no		
XML Address Type				
address1	xs:string	no	123 Main Street	Address line 1
address2	xs:string	no	Suite 1	Address line 2
address3	xs:string	no		Address line 3
City	xs:string	no	Huntersville	
State	xs:string	no	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	no	12345	5-digit U.S. zip code
plus4	xs:string	no	1234	U.S. zip code extension. Ex, 20001-**0001**
XML LineItem Type				
cartLineNumber	xs:int	No	1	This is a number given consecutively to each product in the cart. See Cart Line Number
ATDProductNumber	xs:string	yes	702524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
Quantity	xs:int	yes	1	Quantity of product requested

fillKill	xs:string	no	fok	FOK will kill the order line item if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally and remove unavailable quantity from the order-line item. 'Fill and Kill' or 'Fill or Kill' order line item. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
quickShip	xs:string	no	Y	Quick ship option. See Quick Ship
localPlus	xs:string	No	Y	LocalPlus option. See Local Plus

4.4.13 Request Keys

Table 31: placeOrder Consumer Attribute keys

Key	Value Type	Required	Example	Description
consumerAttribute1	xs:string	No	Any data	Any data specified here < 256 characters
consumerAttribute2	xs:string	No	More of my data	Any data specified here < 256 characters
consumerAttribute3	xs:string	no	And more	Any data specified here < 256 characters

4.4.14 Response Parameters

See the Sample XML Payload Response for xml formatting

Table 32: placeOrder response parameters

Parameter Name	Type	Sample / Format	Description
Order	Order Type		
XML Order type			
orderNumber	xs:string	ATD00943291	ATD-generated cart ID. Read here for more information.
customerPONumber	xs:string	myProdNum	Customer-specified order number passed in request
fillKill	xs:string	fok	FOK will kill the order if the requested quantity of items are not available locally. FAK will fill any of the requested quantity that is available locally, and remove unavailable quantity from the order. 'Fill and Kill' or 'Fill or Kill' order line item. See 'Fill and Kill' / 'Fill or Kill' Allowable values: • fok • fak
Pickup	xs:string	Y	Whether or not the order will be picked up. Allowable values: • Y • N
orderTotal	xs:string	54.46	Total amount of order.
orderLines	OrderLine type		
consumerData	ConsumerData type		Consumer information (name, address, etc.)

XML OrderLine Type			
cartLineNumber	xs:int	1	This is a number given consecutively to each product in the cart. See Cart Line Number
billTo	xs:string	Generic Tire Co.	Name of person or entity to bill the order to
Quantity	xs:int	1	Quantity of item requested.
lineTotal	xs:string	54.46	Line item total price
ATDProductNumber	xs:string	382524825	This is a unique, ATD specific part number. It can contain alphanumeric characters and a period '.'
Cost	xs:string	54.46	Price of item excluding tax in USD.
Fet	xs:string	0.01	Federal Excise Tax amount.
description	xs:string	P215/60R16 Goodride SP06	Product item description
quickShip	xs:string	Y	Quick ship option. See Quick Ship Values: • Y • N
localPlus	xs:string	Y	LocalPlus option. See LocalPlus Values: • Y • N
fulfillments	Fulfillment type		Fulfillment information for line item. If quantity > 1, more than one fulfillment type may be specified depending on fillKill value.
XML Fulfillment Type			
status	xs:string	Filled	Fulfillment status for order. Valid values are: • Filled • Killed

			• Backordered
quantity	xs:int	1	Quantity of items in fulfillment. Quantity <= quantity ordered.
freight	xs:decimal	25.00	Cost of freight (if any) associated with fulfillment.
shipMethod	xs:string	FEDEX	Shipping method of fulfillment.
estimatedDelivery	xs:dateTime	2013-01-01T08:00:00-5.00GMT	Approximate fulfillment delivery date. With timezone
XML ConsumerData Type			
consumerShipTo	ConsumerInfo Type		Ship to details
consumerBillTo	ConsumerInfo Type		Bill to details
consumerAttributes	criteriaType		**NOTE**: Can only be used with prior coordination with ATD.
XML ConsumerInfo Type			
name	xs:string	John Doe	
phoneNumber	xs:string	123-123-1234	
emailAddress	xs:string	j.doe@domain.com	
address	Address Type		
XML Address Type			
address1	xs:string	123 Main Street	Address line 1
address2	xs:string	Suite 1	Address line 2
address3	xs:string		Address line 3
city	xs:string	Huntersville	
state	xs:string	NC	2-character U.S State code. Ex. CA
zipCode	xs:string	12345	5-digit U.S. zip code
plus4	xs:string	1234	U.S. zip code extension. Ex, 20001-**0001**

4.4.15 Error Messages

Table 33: placeOrder error messages

Error Condition	Error Message
Invalid CustomerPONumber	Invalid PO number used for this request
Invalid location number	Please enter a valid location
Internal Order Error	An internal error occurred. Please contact customer support if this error persists
Invalid ATDProductNumbers used in order	Your request did not return any ATD products
Invalid OrderType value specified	OrderType Unknown, OrderType must be D or R
Invalid FillKill value specified	FillKill Unknown, FillKill must be FOK or FAK
Invalid Pickup value specified	Pickup Unknown, Pickup must be Y or N
Item quantity not specified	Please enter a quantity
Invalid QuickShip value	QuickShip Unknown, QuickShip must be Y or N
Incorrect Quickship usage	Quickship is not available for orders using Pickup
Incorrect FillKill usage	FillKill on Header and Lines, FillKill may not be used on both the header and lines
Invalid FillKill value	FillKill Unknown, FillKill must be FOK or FAK

4.4.16 Sample XML Payload

(1) Example Place Order

- **Request**

```
<placeOrderRequest>

  <locationNumber>434471</locationNumber>
  <customerPoNumber>ABC123</customerPoNumber>
  <orderType>D</orderType>
  <pickup>N</pickup>

  <fillKill/>

  <lineitems>
    <lineitem>
      <ATDProductNumber>187000136</ATDProductNumber>
      <quantity>4</quantity>
    </lineitem>
    <lineitem>
      <ATDProductNumber>142000487</ATDProductNumber>
      <quantity>2</quantity>
    </lineitem>
  </lineitems>
</placeOrderRequest>
```

- **Response**

```
<orderResponse>

  <order>
    <orderNumber>ATD123456789</orderNumber>   <=not present in previewOrder
    <customerPoNumber>ABC123</customerPoNumber>
    <pickup>N</pickup>
    <orderTotal>515.10</orderTotal>

    <orderlines>
      <orderline>
        <cartLineNumber>1</cartLineNumber>
        <billTo>House Account</billTo>
        <lineTotal>307.40</lineTotal>
        <quantity>4</quantity >
        <ATDProductNumber>187000136</ATDProductNumber>
        <price>76.85</price>
        <fet>0.00</fet>
        <Description>P215/60R16 Goodride SP06</Description>
        <fulfillments>
          <fulfillment>
            <status>filled</status>
            <quantity>4</quantity>
            <freight>0</freight>
```

```
            <estimatedDelivery>12/29/2012T08:21
            </estimatedDelivery>
          </fulfillment>
        </fulfillments>
      </orderline>
      <orderline>
        ...
        <cartLineNumber>2</cartLineNumber>
        <billTo>House Account</billTo>
        <lineTotal>207.70</lineTotal>
        <ATDProductNumber>142000487</ATDProductNumber>
        ...
        <fulfillments>
          <fulfillment>
            <status>Backordered</status>
            ...
           </fulfillment>
        </fulfillments>

      </orderline>
    </orderlines>
  </order>
</orderResponse>
```

(2) Example Order level Fill or Kill

This demonstrates Fill or Kill at the Order Level. In the example, the setting *fok* kills all items in the order because one of the items is unavailable, having a fulfillment status of *backordered*

- **Request**

```
<placeOrderRequest>

  <locationNumber>434471</locationNumber>
  ...
  <fillKill>fok</fillKill>  <= tells ATD to kill the whole order if any
                          one of the items cannot be filled
  <lineitems>
    <lineitem>
      <ATDProductNumber>187000136</ATDProductNumber>
      <quantity>4</quantity>
    </lineitem>
   <lineitem>
      <ATDProductNumber>142000487</ATDProductNumber>
      <quantity>2</quantity>
    </lineitem>

  </lineitems>
</placeOrderRequest>
```

- **Response**

```
<orderResponse>

  <order>

    <orderNumber>ATD123456789</orderNumber>   <=not present in previewOrder
    <!--<orderTotal/>--> <= no total. Killed items
    ...
    <orderlines>
      <orderline>
        ...
        <!--<lineTotal/>--> <= no total. Killed item
        <ATDProductNumber>187000136</ATDProductNumber>
        ...
        <fulfillments>
          <fulfillment>
            <status>killed</status>  <= filled item now killed
            ...
          </fulfillment>
        </fulfillments>
      </orderline>
      <orderline>
        ...
        <!--<lineTotal/>--> <= no total. Killed item
        <ATDProductNumber>142000487</ATDProductNumber>
        ...
        <fulfillments>
          <fulfillment>
            <status>killed</status> <= backordered item triggered the
                                  whole order to be killed
            ...
           </fulfillment>
        </fulfillments>

      </orderline>
    </orderlines>
  </order>
</orderResponse>
```

(3) <u>Example Line-item level Fill and Kill</u>

Fill and Kill (*fak)* tells ATD to fill available items and kill all others. This example illustrates this behavior at the line level. Backordered items are automatically killed, and available items filled.

- **Request**

```
<placeOrderRequest>
  ...
  <lineitems>
    <lineitem>
      <ATDProductNumber>187000136</ATDProductNumber>
      <quantity>4</quantity>
      <fillKill>fak</fillKill>  <= line level fillKill
    </lineitem>
   <lineitem>
      <ATDProductNumber>142000487</ATDProductNumber>
      <quantity>2</quantity>
      <fillKill>fak</fillKill> <= line level fillKill
    </lineitem>

  </lineitems>
</placeOrderRequest>
```

- **Response**

```
<orderResponse>

 <order>
   ...
   <orderTotal>411.25</orderTotal>  <= 5 items filled
   <orderlines>
     <orderline>
       ...
       <lineTotal>307.40</lineTotal>
       <ATDProductNumber>187000136</ATDProductNumber>
       ...
       <fulfillments>
         <fulfillment>
           <status>filled</status> <= item filled
           <quantity>4</quantity>
           ...
         </fulfillment>
       </fulfillments>
     </orderline>
     <orderline>
       ...
       <lineTotal>103.85</lineTotal> <= total updated. Only 1 item killed
       <ATDProductNumber>142000487</ATDProductNumber>
       ...
       <fulfillments>
```

```
          <fulfillment>
            <status>killed</status> <= item killed. Was backordered
            <quantity>1</quantity>
            ...
          </fulfillment>

          <fulfillment>
            <status>filled</status> <= item filled 1 available
            <quantity>1</quantity>
            ...
          </fulfillment>

        </fulfillments>

      </orderline>
    </orderlines>
  </order>
</orderResponse>
```

5. ORDER STATUS SERVICE

5.1 Description:

The ATDConnect Order Status service provides a way to check the status of orders that have already been placed.

It consists of two functions: getOrderStatusByCriteria and getOrderDetail

The getOrderStatusByCriteria function retrieves order-level information on orders that have already been placed, based on user-specified search criteria.

The getOrderDetail function retrieves full order details on orders that have already been placed.

5.2 WSDL Location

Production:	https://ws.atdconnect.com/ws/3_2/orderStatus.wsdl
Sandbox/Test:	https://testws.atdconnect.com/ws/3_2/orderStatus.wsdl

5.3 Service Method Calls

5.3.1 View Order Status (getOrderStatusByCriteria)

5.3.2 Function Name

getOrderStatusByCriteria

5.3.3 Description:

This call retrieves basic header information for orders that have already been placed.

<u>Please read</u> the Functional Considerations section below to understand nuances of this function's operation.

5.3.4 Functional Considerations

When using a date range as your search criteria, there are a few things to take into consideration:

- This call searches the past 90 days of orders by default if a custom Start Date and End Date is not supplied.
- If supplying your own custom dates, it can be any 90 day range but it must not exceed 90 days total.

- If your custom dates exceed a 90 day window, an error will be returned.
- Start Date and End Date are not required criteria but the other must be supplied if you supply one.

5.3.5 Request Parameters

Table 34: getOrderStatusByCriteria request parameters

Parameter Name	Type	Required	Sample / Format	Description
locationNumber	string	Yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location
criteria	Criteria Type	no	See Table 35 for valid key options	

5.3.6 Request Keys

Table 35: getOrderStatusByCriteria request criteria keys

Key	Value Type	Required	Example	Description
Confirmation Number	string	No	ATD01234567	ATD-generated cart ID. It can contain multiple order numbers. See section 5.3.14 for more.
Customer Po Number	string	No	123456789	This is a non-unique, customer-provided value. It can contain a combination of alphanumeric characters.
Start Date	date	No	2015-12-10	Format must be YYYY-MM-DD. End Date must also be supplied. See section 5.3.4 for more.
End Date	date	No	2015-12-10	Format must be YYYY-MM-DD. Start Date must also be supplied. See section 5.3.4 for more.

5.3.7 Response Parameters

Table 36: getOrderStatusByCriteria response parameters

Parameter Name	Type	Sample / Format	Description
orderStatuses	Parent type for Order Status Type collection		Collection of order statuses returned in search
XML Order Status Type			
confirmationNumber	string	ATD01234567	ATD-generated cart ID. It can contain multiple order numbers. See section 5.3.14 for more.
customerPoNumber	string	123456789	This is a non-unique, customer-provided value. It can contain a combination of alphanumeric characters.
orderTotal	string	54.46	Total amount of order.
datePlaced	dateTime	2015-12-17T14:30:03-05:00	Date format is YYYY-MM-DD. Time format is as follows: (UTC time the order was placed) – (how many hours the dealer timezone is behind UTC time) = the actual time the dealer placed the order
status	string	Open	Status for the shopping cart as a whole. Valid values are: • Open • Closed
orderNumbers	Parent type for Order Number Type collection		Collection of order numbers returned in search.

invoiceNumbers	Parent type for Invoice Number Type collection		Collection of invoice numbers returned in search.
categoryCode	string	ORDER	
XML Order Number Type			
orderNumber	string	93283999	The order number assigned to a specific shipment going to the dealer.
XML Invoice Number Type			
invoiceNumber	string	S069987709	The invoice number assigned to a specific shipment going to the dealer.

5.3.8 Response Keys

No name-value pairs for this function exist

5.3.9 Error Messages

Table 37: getOrderStatusByCriteria error messages

Error Condition	Error Message
The criteria type entered as a key is invalid	Invalid Criteria key: <entered search criteria key>
The date search values do not match the valid format	Both Start Date and End Date criteria values must be entered
The date search range only provides one criteria without providing the other	Both Start Date and End Date criteria values must be entered
The date search criteria End Date value	End Date search criteria cannot be before the default start date

is before the Start Date value	
The date range entered exceeds 90 days	Date Range searches cannot exceed 90 days of range

5.3.10 Sample XML Payload

(1) Order Status using Confirmation Number as Criteria

- **Request**


```
<getOrderStatusByCriteriaRequest>
      <locationNumber>434471</locationNumber>
      <criteria>
        <entry>
          <key>Confirmation Number</key>
          <value>ATD01234567</value>
        </entry>
      </criteria>
</getOrderStatusByCriteriaRequest>
```


- **Response**


```
<orderStatusCriteriaResponse>
  <orderStatuses>
    <orderStatus>
      <confirmationNumber>ATD01234567</confirmationNumber>
      <customerPoNumber>201510091234</customerPoNumber>
      <orderTotal>132.38</orderTotal>
      <datePlaced>2015-10-09T10:22:28-05:00</datePlaced>
      <status>Open</status>
      <orderNumbers>
        <orderNumber>95552799</orderNumber>
      </orderNumbers>
      <invoiceNumbers>
        <invoiceNumber>S099994687</invoiceNumber>
      </invoiceNumbers>
      <categoryCode>ORDER</categoryCode>
    </orderStatus>
  </orderStatuses>
</orderStatusCriteriaResponse>
```

(2) Order Status using PO Number as Criteria

- **Request**


```
<getOrderStatusByCriteriaRequest>
        <locationNumber>434471</locationNumber>
        <criteria>
          <entry>
            <key>Customer Po Number</key>
            <value>201510091234</value>
          </entry>
        </criteria>
</getOrderStatusByCriteriaRequest>
```


- **Response**


```
<orderStatusCriteriaResponse>
  <orderStatuses>
    <orderStatus>
      <confirmationNumber>ATD01234567</confirmationNumber>
      <customerPoNumber>201510091234</customerPoNumber>
      <orderTotal>132.38</orderTotal>
      <datePlaced>2015-10-09T10:22:28-05:00</datePlaced>
      ...
</orderStatusCriteriaResponse>
```


(3) Order Status using Date Range as Criteria

- **Request**


```
<getOrderStatusByCriteriaRequest>
        <locationNumber>434471</locationNumber>
        <criteria>
          <entry>
```

```
      <key>Start Date</key>
      <value>2016-01-12</value>
    </entry>
    <entry>
      <key>End Date</key>
      <value>2016-01-18</value>
    </entry>
  </criteria>
</getOrderStatusByCriteriaRequest>
```

Start Date should be the date furthest from the current calendar date
End Date should be the date closest to the current calendar date

- **Response**

```
<orderStatusCriteriaResponse>
    <orderStatuses>
      <orderStatus>
        <confirmationNumber>ATD91234867</confirmationNumber>
        <customerPoNumber>2015011812340002</customerPoNumber>
        <orderTotal>132.04</orderTotal>
        <datePlaced>2016-01-18T11:05:55-05:00</datePlaced>
        …
      </orderStatus>
      <orderStatus>
        <confirmationNumber>ATD91224364</confirmationNumber>
        <customerPoNumber>201501120001</customerPoNumber>
        <orderTotal>2030.08</orderTotal>
        <datePlaced>2016-01-12T10:59:47-05:00</datePlaced>
        …
      </orderStatus>
    <orderStatuses>
<orderStatusCriteriaResponse>
```

5.3.11 View Order Status Details (getOrderDetail)

5.3.12 Function Name

getOrderDetail

5.3.13 Description:

This call retrieves full order details for orders that have already been placed.

Please read the Functional Considerations section below to understand nuances of this function's operation.

5.3.14 Functional Considerations

The user must search by Confirmation Number. Only one Confirmation Number is allowed per search.

The Confirmation Number is considered a shopping cart to ATD and can contain multiple Order Numbers within it. This is because multiple products can be ordered in the same placeOrder request and each one of those products may be filled from a different Distribution Center, depending on their availability. Therefore, products in the same shopping cart that get fulfilled by a different Distribution Center will have a different Order Number so they can be tracked and invoiced appropriately. *Currently, the orderNumber response parameter in the placeOrder function is the same as Confirmation Number in the Order Status service.*

5.3.15 Request Parameters

Table 38: getOrderDetail request parameters

Parameter Name	Type	Required	Sample / Format	Description
locationNumber	string	Yes	434471	A unique identifier used by ATD to distinguish customers. It is specific to a single shipping location
confirmationNumber	string	Yes	ATD01234567	ATD-generated cart ID. It can contain multiple order numbers. See section 5.3.14 for more.

5.3.16 Request Keys

No name-value pairs for this function exist

5.3.17 Response Parameters

Table 39: getOrderDetail response parameters

Parameter Name	Type	Sample / Format	Description

orderDetail	Parent type for Order Detail Type collection		Collection of order details returned in search
XML Order Detail Type			
confirmationNumbe r	string	ATD01234567	ATD-generated cart ID. It can contain multiple order numbers. See section 5.3.14 for more.
customerPoNumber	string	20335	This is a non-unique, customer-provided value. It can contain a combination of alphanumeric characters.
datePlaced	dateTime	2015-12-17T14:30:03-05:00	Date format is YYYY-MM-DD. Time format is as follows: (UTC time the order was placed) – (how many hours the dealer location is behind UTC time) = the actual time the dealer placed the order
orderTotal	string	2529.29	Total amount of order in USD, including taxes.
taxes	string	313.49	Tax amount on total order.
pickup	string	Delivery	Whether or not the order will be picked up. Allowable values: • Y • N
createdBy	string	TIRE COMPANY, LTD.	Customer name.
approvedBy	string	Smith, John	ATD sales rep attached to the customer's account.
customerComments	string	Hold order for John Smith	Comments added by customer on ATDOnline.

orderLines	parent type for Order Line Type collection		Collection of all line items ordered in the shopping cart.
XML Order Line Type			
billTo	string	Generic Tire Co.	Name of person or entity to bill the order to
cartLineNumber	xs:int	1	This is a number given consecutively to each product in the cart. See Cart Line Number
quantity	int	1	Quantity of all items ordered in the shopping cart.
atdProductNumber	string	21201	This is a unique, ATD-specific part number. It can contain alphanumeric characters and a period '.'
cost	decimal	199.20	Price of individual item excluding tax in USD.
fet	decimal	0.01	Federal Excise Tax amount.
description	string	P235/55R17 98H MIC PILOT MXM4 BW	Product item description
fulfillments	Parent type for Fulfillment Type collection		Collection of all order numbers in the shopping cart.
XML Fulfillment Type			
status	string	Shipped	Status for the order. Valid values are: • At DC • Delivered
quantity	int	1	Quantity of a specific item ordered in the shopping cart.

lineTotal	decimal	942.00	Individual cost of the product that was ordered multiplied by the quantity ordered.
shipMethod	string	ATD Route Truck	Shipping method of fulfillment.
estimatedDeliveryDate	dateTime	2015-12-02T00:00:00.000-05:00	Approximate fulfillment delivery date with timezone.
actualDeliveryDate	dateTime	2015-12-02T00:00:00.000-05:00	Actual fulfillment delivery date with timezone.
orderNumber	string	83283414	ATD internal order number.
trackingNumber	string	740011799223115740791	Carrier tracking number.
trackingUrl	string	https://www.fedex.com/apps/fedextrack/index.html	Carrier's tracking website.
invoiceNumber	string	S068284702	ATD internal invoice number.
deliveryReceiptNumber	string	OS123456	The receipt for national account claims. This will always display as an empty field.
quantityReturnable	int	4	Quantity that is returnable to the DC.
quantityBackordered	int	1	Quantity that has been placed on backorder.
quantityShipped	int	4	Quantity that has been shipped.
sourceType	string	LOCAL	The region the order was sourced from. Valid values are • LOCAL • LOCALPLUS • NATIONAL
sourceTypeDC	string	INDIANAPOLIS, IN	DC the order was sourced from.

5.3.18 Response Keys

No name-value pairs for this function exist

5.3.19 Error Messages

No returnable errors.

5.3.20 Sample XML Payload

- **Request**


```xml
<getOrderDetailRequest>
        <locationNumber>434471</locationNumber>
        <confirmationNumber>ATD09777394</confirmationNumber>
</getOrderDetailRequest>
```


- **Response**


```xml
<orderStatusDetailResponse>
   <orderDetail>
              <confirmationNumber>ATD09777394</confirmationNumber>
              <customerPoNumber>201510091234</customerPoNumber>
              <orderTotal>141.88</orderTotal>
              <datePlaced>2015-10-09T10:22:28-05:00</datePlaced>
              <taxes>9.28</taxes>
              <pickup>Delivery</pickup>
              <createdBy>Tire Shop USA, Inc.</createdBy>
              <approvedBy>Smith, John</approvedBy>
              <customerComments>Hold order for John Smith</customerComments>
              <orderLines>
                 <orderLine>
                            <billTo>TIRE SHOP-NATL</billTo>
                            <cartLineNumber>1</cartLineNumber>
                            <quantity>1</quantity>
                            <atdProductNumber>29901</atdProductNumber>
                            <cost>132.6</cost>
                            <fet>0.00</fet>
                            <description>235/55ZR17 99W BFG g-FORCE COMP 2 A/S CPJ</description>
                             <fulfillments>
                               <fulfillment>
                                  <status>Delivered</status>
                                  <quantity>1</quantity>
                                  <lineTotal>132.6</lineTotal>
                                  <shipMethod>FEDEX Ground</shipMethod>
```

```
                    <orderNumber>95552799</orderNumber>
                    <trackingNumber>9374889879004610499999</trackingNumber>
                    <trackingUrl>https://www.fedex.com/apps/fedextrack/index.html
                    </trackingUrl>
                    <invoiceNumber>S099994488</invoiceNumber>
                    <deliveryReceiptNumber/>
                    <quantityReturnable>0</quantityReturnable>
                    <quantityBackordered>0</quantityBackordered>
                    <quantityShipped>1</quantityShipped>
                    <sourceType>LOCAL</sourceType>
                    <sourceTypeDC>CHARLOTTE</sourceTypeDC>
                  </fulfillment>
                </fulfillments>
          </orderLine>
        </orderLines>
  </orderDetail>
</orderStatusDetailResponse>
```

APPENDIX 1: ATDCONNECT ACCESS AND SUPPORT

Environments

ATDConnect provides both production and sandbox/test environments.
Authorized clients may access the sandbox environment with a valid username/password and a clientID issued by ATD.
Access to the production environment is pending until joint testing and mutual sign off is obtained between ATD and the client.

Sample Code

ATDConnect has been developed and successfully deployed in a variety of application environments. ATD can provide sample code upon request.

Product Questions?

To obtain Client IDs, request web service enhancements or provide feedback on ATDConnect, please contact:

Brock Bowman, Director of Sales Support

Phone: 704-632-7038

Email: bbowman@atd-us.com

Technical Support

For technical questions or to report operational problems, please contact our Customer Care team. They can be reached at

Phone: 1-800-511-1226

Email: online_support@atd-us.com

APPENDIX 2: PRODUCT SEARCH TIRE AND WHEEL SPECIFICATIONS

1. Ply Rating

Load Range	Ply Rating	Load Range	Ply Rating
A	2	G	14
B	4	H	16
C	6	J	18
D	8	L	20
E	10	M	22
F	12	N	24

2. Speed Rating Chart

Speed Symbol	Speed (km/h)	Speed (mph)
B	50	31
C	60	37
D	65	40
E	70	43
F	80	50
G	90	56
J	100	62
K	110	68
L	120	75
M	130	81

N	140	87
P	150	94
Q	160	100
R	170	106
S	180	112
T	190	118
U	200	124
H	210	130
V	240	149
W	270	168
Y	300	186
(Y)	Above 300	Above 186

3. Bolt Circle/Pattern Chart

Bolt Circle Conversions - Common ones are in **bold**

mm	inches
4 x 100	4 x 3.94
5 x 100	5 x 3.94
4 x 108	**4 x 4.25**
5 x 108	**5 x 4.25**
4 x 110	4 x 4.33
5 x 110	5 x 4.33
5 x 112	5 x 4.41
4 x 114.3	**4 x 4.5**
5 x 114.3	**5 x 4.5**

6 x 114.3	**6 x 4.5**
5 x 115	5 x 4.52
5 x 120	5 x 4.72
5 x 120.7	**5 x 4.75**
5 x 127	**5 x 5**
6 x 127	**6 x 5**
5 x 130	5 x 5.12
5 x 135	5 x 5.3
6 x 135	6 x 5.3
5 x 139.7	**5 x 5.5**
6 x 139.7	**6 x 5.5**
8 x 165.1	**8 x 6.5**
8 x 170	8 x 6.69

4. ATD Finish Values

Anthracite	Machined Finished
Black	Opal
Blue	Orange
Bronze	Polished
Chrome	Red
Galvanized	Silver
Gold	Two-Tone White
Green	

5. Tire Sidewall Chart

Sidewall Description	Sidewall Description
Black Circumference Serrated Letter	Raised Ribbed Black Letter
Black Letter	Raised Serrated Band
Black Serrated Letter	Raised White
Black Wall	Raised White Letter
Broken Serrated Band	Recessed Black
Broken Serrated Band - Emt	Serrated Black Letter
Diagonal Serrated Band	Solid Red Letter
Heavy Scruff Rib Black	Vertical Serrated Band
Outlined Raised Black Letter	Vertical Serrated Undulating Band - Black Solid
Outlined Raised White Letter	Vertical Serrated Undulating Unevenly Broken Band
Outlined White Letter	White Wall
Raised Black Letter	Xtra Narrow White Stripe
Raised Outlined White Letter	